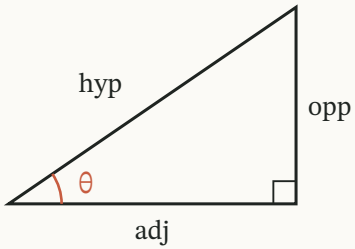
Trigonometry CHEAT SHEET

THE SIX RATIOS



$\sin\theta = \frac{\text{opp}}{\text{hyp}}$ $\quad \text{cosec}\,\theta = \frac{1}{\sin\theta} = \frac{\text{hyp}}{\text{opp}}$

$\cos\theta = \frac{\text{adj}}{\text{hyp}}$ $\quad \sec\theta = \frac{1}{\cos\theta} = \frac{\text{hyp}}{\text{adj}}$

$\tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{\sin\theta}{\cos\theta}$ $\quad \cot\theta = \frac{1}{\tan\theta} = \frac{\text{adj}}{\text{opp}}$

cosec pairs with sin · sec with cos · cot with tan

SPECIAL ANGLES

θ		sin θ	cos θ	tan θ
0°	0	0	1	0
30°	π/6	1/2	√3/2	1/√3
45°	π/4	√2/2	√2/2	1
60°	π/3	√3/2	1/2	√3
90°	π/2	1	0	undefined

sin θ: √0/2, √1/2, √2/2, √3/2, √4/2 (cos θ: the same, reversed)

$\sin(90° - \theta) = \cos\theta$ $\quad \tan(90° - \theta) = \cot\theta$

ANGLES AND RADIANS

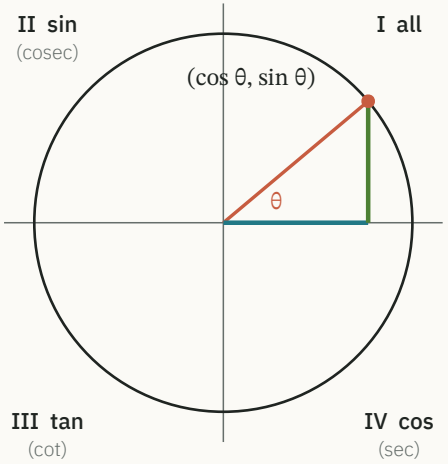
π radians $= 180°$ $\quad 1$ radian $\approx 57.3°$

degrees → radians: $\times \frac{\pi}{180°}$

arc length $l = r\theta$ $\quad$ sector area $= \frac{1}{2}r^2\theta$

θ in radians · anticlockwise is positive

THE UNIT CIRCLE



corner labels: the ratios positive there

reference angle
for θ in II: $180° - \theta$
in III: $\theta - 180°$
in IV: $360° - \theta$

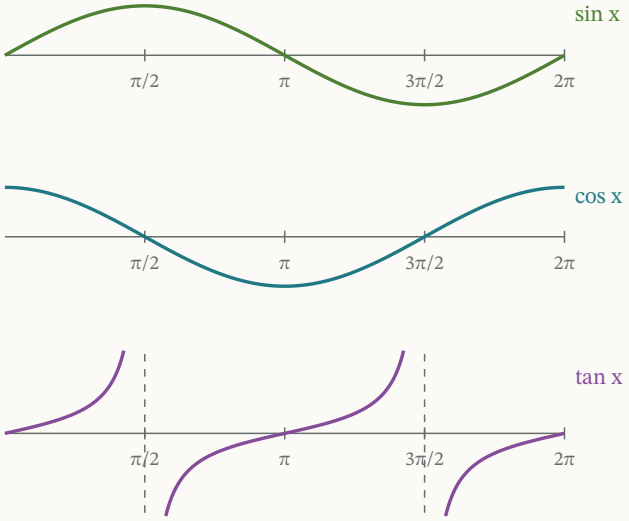
$\tan\theta = y / x$
$-1 \le \sin\theta, \cos\theta \le 1$

the point at angle θ: across = cos θ, up = sin θ

PYTHAGORAS ON THE CIRCLE

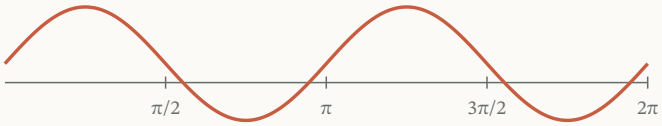
$\sin^2\theta + \cos^2\theta = 1$
$1 + \tan^2\theta = \sec^2\theta$
$1 + \cot^2\theta = \text{cosec}^2\theta$

GRAPHS



period: sin, cos, cosec, sec 2π · tan, cot π
$\cos(-x) = \cos x$ · $\sin(-x) = -\sin x$

STRETCH, SQUEEZE, LIFT



$y = a\sin(bx) + c$

amplitude $|a|$ $\quad$ period $\frac{2\pi}{|b|}$ $\quad$ moved up by c

drawn: a = 1.5, b = 2, c = 0.5 (period π)

Trigonometry CHEAT SHEET

IDENTITIES

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \operatorname{cosec}^2\theta$

COMPOUND ANGLES

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\sin(A - B) = \sin A \cos B - \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

sin: the sign between matches the sign inside · cos: it flips

ALLIED ANGLES

180°, 360°: same function · 90°, 270°: sin ↔ cos, tan ↔ cot

sign: the original function's sign where the angle lands (x acute)

angle		sin	cos	tan
$-x$		$-\sin x$	$\cos x$	$-\tan x$
$90° - x$	$\pi/2 - x$	$\cos x$	$\sin x$	$\cot x$
$90° + x$	$\pi/2 + x$	$\cos x$	$-\sin x$	$-\cot x$
$180° - x$	$\pi - x$	$\sin x$	$-\cos x$	$-\tan x$
$180° + x$	$\pi + x$	$-\sin x$	$-\cos x$	$\tan x$
$270° - x$	$3\pi/2 - x$	$-\cos x$	$-\sin x$	$\cot x$
$270° + x$	$3\pi/2 + x$	$-\cos x$	$\sin x$	$-\cot x$
$360° - x$	$2\pi - x$	$-\sin x$	$\cos x$	$-\tan x$

DOUBLE ANGLES

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$\quad = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

HALF ANGLES

$\cos^2 A = \frac{1 + \cos 2A}{2} \qquad \sin^2 A = \frac{1 - \cos 2A}{2}$

$\tan \frac{\theta}{2} = \frac{\sin \theta}{1 + \cos \theta}$

TRIPLE ANGLES

$\sin 3A = 3 \sin A - 4 \sin^3 A$

$\cos 3A = 4 \cos^3 A - 3 \cos A$

$\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$

PRODUCT TO SUM

$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$

$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$

$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$

$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

SUM TO PRODUCT

$\sin C + \sin D = 2 \sin \frac{C + D}{2} \cos \frac{C - D}{2}$

$\sin C - \sin D = 2 \cos \frac{C + D}{2} \sin \frac{C - D}{2}$

$\cos C + \cos D = 2 \cos \frac{C + D}{2} \cos \frac{C - D}{2}$

$\cos C - \cos D = -2 \sin \frac{C + D}{2} \sin \frac{C - D}{2}$

GENERAL SOLUTIONS

n is any integer

$\sin x = \sin \alpha \Rightarrow x = n\pi + (-1)^n \alpha$

$\cos x = \cos \alpha \Rightarrow x = 2n\pi \pm \alpha$

$\tan x = \tan \alpha \Rightarrow x = n\pi + \alpha$

INVERSE FUNCTIONS

	domain	principal value
$\sin^{-1}x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}x$	all x	$(-\pi/2, \pi/2)$
$\operatorname{cosec}^{-1}x$	$\lvert x\rvert \geq 1$	$[-\pi/2, \pi/2]$, not 0
$\sec^{-1}x$	$\lvert x\rvert \geq 1$	$[0, \pi]$, not $\pi/2$
$\cot^{-1}x$	all x	$(0, \pi)$

$\sin^{-1}x$ is an angle, not 1 / sin x

ANY TRIANGLE

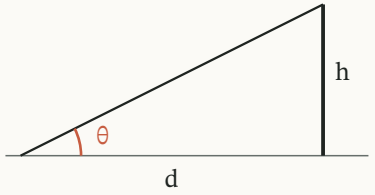
sine rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$

$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \qquad \text{area} = \frac{1}{2}ab \sin C$

two sides and a non-included angle can give two triangles

HEIGHTS AND DISTANCES



$h = d \tan \theta$

depression from above
= elevation from below